MIL-P-46499 (Ord)
3 March 1961

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M78
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Percussion, M78.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-2550 - General Specification for Ammunition Except Small Arms Ammunition.
JAN-P-223 - Powder, Black.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

7645336 - Primer, Caliber .45
8847520 - Primer, Percussion, M78
8850251 - Box, Packing, Ammunition, for Primer Percussion, M78
8850252 - Box, Set-up, Packing Ammunition for Primer, Percussion, M78

FSC 1390

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(see 6.2) - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer.-The primer shall comply with all requirements specified on Drawing (dwg.) 8847520 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.7 percent when determined as specified in 4.4.1.

3.4 Sensitivity.

3.4.1 Non-functioning.-The primer shall fail to function under an applied energy value of 5.76 inch ounces minimum (min.), when tested as specified in 4.4.2.1.

3.4.2 Functioning.-The primer shall function under an applied energy value of 29.40 inch ounces maximum (max.), when tested as specified in 4.4.2.2.

3.5 Waterproofness and security of parts.-The primers shall function after being immersed in water to a min. depth of 2 inches for 48 hours, without evidence of hangfire, blowback, or separation of parts when tested as specified in 4.4.3.

3.6 Workmanship.-All parts shall be free from chips, dirt, grease and other foreign matter. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for the control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.3 and 4.4 and the other documents referenced herein. Except in the cases of 4.4.2.1 and 4.4.2.2, the contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein. In the case of the non-functioning and functioning tests of the primer, the weight of the ball and height of drop shall not differ from the values specified in 4.4.2.1 and 4.4.2.2 and on the applicable test equipment drawing.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

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e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference, with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

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4.2 Initial production testing.-The first lot produced in accordance with 4.3.1, and all successive lots, prior to Government acceptance of the initial production quantity shall be inspected as required in 4.3.2 and 4.3.3 of this specification, except that the sampling plans for the sensitivity and waterproofness tests shall be as specified below. The contractor shall continue these plans of inspection until three consecutive lots have complied with the requirements. The samples for the initial production tests shall be selected at random by the Government inspector.

4.2.1 Sensitivity (see 3.4).

4.2.1.1 Non-functioning (see 3.4.1) Critical defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IVA, Code letter L, and an AQL of 0.065 percent. The test shall be as specified in 4.4.2.1, using test equipment specified in 4.3.5.

4.2.1.2 Functioning (see 3.4.2) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter O and an AQL of 0.15 percent. The test shall be performed as specified in 4.4.2.2, using test equipment specified in 4.3.5.

4.2.1.3 Waterproofness and security of parts (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter L, and an AQL of 1.00 percent. The test shall be as specified in 4.4.3 using equipment in accordance with 4.3.5.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of primer assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.

b. Black powder of one interfix lot number from one manufacturer.

c. Primer dwg. B 7645336 (Primer element-see 6.4) of one lot number from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.5.

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4.3.2.1 Primer, prior to assembling holder into head and closure to holder (see dwg. C-8847520).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Weight of charge	Balance
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.6)	Visual

4.3.2.2 Primer, prior to assembling holder into head (see dwg. C-8847520).

Categories	Defects	Method of inspection
Critical:		
1.	Primer element above flush in holder	Straight edge
2.	Length from forward end of head to top of firing plug, min	Gage
Major:	AQL 0.25 percent	
101.	Primer element missing, improperly sealed or loose	Visual-Manual
102.	Lacquer on face of primer element	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.6)	Visual

4.3.2.3 Assembly (see dwg. C-8847520).

Categories	Defects	Method of inspection
Critical:		
1.	Firing plug above flush with surface of head	Straight edge
Major:	AQL 0.40 percent	
101.	Holder fails disassembly torque	Torque-wrench
102.	Seal of closure missing or incomplete	Visual
103.	Closure punctured	Visual
104.	Firing plug missing	Visual
Minor:	AQL 1.00 percent	
201.	Evidence of poor workmanship (see 3.6)	Visual
202.	Disc wrinkled or crimp incomplete for full 360°	Visual

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4.3.2.4 Box, set-up, packing, prior to sealing (see dwg. 8850252).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.00 percent	
201.	Number of primers incorrect	Visual
202.	Insufficient packing material	Visual

4.3.2.5 Box, set-up, packing (see dwg. 8850252).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Box damaged to the extent that contents are exposed or liable to become exposed during subsequent handling	Visual
102.	Marking misleading or unidentifiable	Visual
Minor:	AQL 1.00 percent	
201.	Contents move when shaken	Manual
202.	Sealing strip incomplete, wrinkled, or not in contact with surface of box for at least 1/2 inch around entire box	Visual

4.3.2.6 Box, packing (see dwg. 8850251).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Box damaged to extent that contents are exposed or liable to become exposed in subsequent handling	Visual
Minor:	AQL 1.00 percent	
201.	Strapping missing, broken, loose, or improperly assembled	Visual-Manual
202.	Marking misleading or unidentifiable	Visual

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4.3.3 Testing.

4.3.3.1 Moisture content of black powder (see 3.3) Major defect.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test for verification at least one completed primer assembly from each eight hours production. Composite samples shall not be used in determining compliance. The black powder shall be tested as specified and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.3.3.2 Sensitivity.

4.3.3.2.1 Non-functioning (see 3.4.1) Critical defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IVA, Code letter I, and AQL of 0.25 percent. The test shall be performed as specified in 4.4.2.1 using equipment in accordance with 4.3.5.

4.3.3.2.2 Functioning (see 3.4.2) Major defect.-A sample of 125 assemblies shall be selected from each lot for this test. If two or more assemblies fail to comply with the requirement, the lot shall be rejected. If only one defective is found, a second sample of 250 assemblies shall be tested. If the combined number of defectives in both the first and second samples is two or more the lot shall be rejected. The test shall be performed as specified in 4.4.2.2 using equipment specified in 4.3.5.

4.3.3.3 Waterproofness and security of parts (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter I, and AQL of 1.00 percent. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.5.

4.3.4 Check test for deterioration of the primer element.-If the total time between the manufacture of any primer element lot and the assembly of that lot into M78 primers exceeds 2 years or if the primer elements have been subjected to adverse conditions, however brief, at any time since previous tests, the primer element lot shall be subjected to and must satisfactorily comply with the check test specified in 4.4.4 immediately before the primer elements are assembled into the M78 primers. The test shall be performed on a sample of 325 primer elements selected by the Government inspector at the facility assembling the primers element into the M78 primers. Twenty-five primer elements shall be tested at each height. If the sum of the mean critical height (see 6.5) $\bar{H}$ plus (/) 5 standard deviations (s) exceeds the upper limit (15 inches), or $\bar{H}$ -2s is less than the lower limit (3 inches) the lot shall be rejected. The test shall be performed as specified in 4.4.4 using equipment specified in 4.3.5.

4.3.5 Inspection equipment.-For the performance of the examinations and tests specified in 4.3 and 4.4, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment (see 6.2). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the

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required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test methods and procedures.

4.4.1 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification JAN-P-223.

4.4.2 Sensitivity.

4.4.2.1 Non-functioning.-The primer, less the black powder charge, shall be assembled in the test fixture specified in 4.3.5 so that the ball will strike the center of the plug. A steel ball weighing 1.94 ± .02 ounces shall be dropped on the plug from a height of three (3) inches min. The height of drop shall be measured as the distance between the bottom of the ball and the top of the plug.

4.4.2.2 Functioning.-The test shall be performed as specified in 4.4.2.1 using the same weight ball except that the drop height shall be fifteen (15) inches max.

4.4.3 Waterproofness and security of parts.-The primer assemblies shall be immersed to the required depth in water at a temperature of 70 ± 5 degrees Fahrenheit (°F.) for the 48 hour period. The primer assemblies shall be removed from the water, wiped dry, assembled in a fixture similar to the one used in 4.4.2.1, and fired by the same method specified in 4.4.2.1. Any primer assemblies that fail to fire shall be disassembled and examined for evidence of leakage. Observation shall be made for evidence of hangfire, blowback and separation of parts.

4.4.4 Check test for deterioration of primer element.-The primer elements shall be stored at a temperature of 75 ± 5°F. for a min. of 24 hours and tested immediately thereafter. The sensitivity testing fixture shall be set at a drop height that is midway between the specified upper and lower limits and 25 primer elements shall be fired. Observation shall be made as to how many primer elements fire, fail to fire, blowback, or hangfire, and this data recorded. This entire procedure shall be repeated at successively increasing one inch heights, testing 25 sample primer elements at each height, until a height is reached at which all 25 primer elements fire. The foregoing procedure shall be repeated, starting with a drop height one inch below the midway starting height and continuing at successively decreasing one inch heights until a height is reached at which all 25 primer elements fail to fire. Primer elements which fail to function at any intermediate height shall be discarded and shall not be used for any of the succeeding heights.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.
5.1.1 Level C.-Packaging shall be in accordance with dwg. 8850252.
5.2 Packing.
5.2.1 Level C.-Packing and marking shall be in accordance with dwg. 8850251.
5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.
6.2 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.
6.3 Check test for deterioration of M78 primer.-The following tests are designed to prevent loading M78 primers which have deteriorated into a major ammunition item. The tests should be performed on each lot of M78 primers which have been stored under normal conditions for more than 2 years, or under adverse conditions for any length of time. If the specification for the ammunition item into which the M78 primers are loaded does not require that the check test be performed, the contracting officer should insert the provision for the check test in the contract.
6.3.1 Non-functioning sensitivity (see 3.4).-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter J, and AQL of .65 percent. The test shall be performed as specified in 4.4.2.1
6.3.2 Functioning sensitivity (see 3.4.2).-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter H, and AQL of .15 percent. The test shall be performed as specified in 4.4.2.2.
6.4 Primer, dwg. B-7645336.-The words "primer element" will hereafter be used in this specification to denote "primer, dwg. B-7645336".
6.5 Mean critical height.-The mean critical height is the height at which 50 percent of the primers fire and 50 percent of the primers misfire.
6.6 Formulae and method of determining $\bar{H}$ and s.
6.6.1 Formulae.

$$\bar{H} = \sum pi \; / \; (H_{100\%} \; / \; .50)$$

$$s = \sqrt{(\sum K_i P_i) - (\sum P_i)^2}$$

where:
$\bar{H}$ = mean critical height, or the height at which 50 percent of the primers fire and 50 percent of the primers misfire.
$\sum$ = sum of the individual values.
P_i = decimal fraction of primers failing to fire at each individual height.

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$H_{100\%}$ + .50 = first height at which all primers in sample fail to fire, plus .50.

s = standard deviation, or dispersion of the individual primer sensitivities about the mean.

K_1 = standard deviation factor.

6.6.2 Method.-The data should be tabulated as follows: Divide a sheet of paper in 7 columns and number the columns I through VII.

Title Column I "Height of Drop" and test all intermediate heights of drop, in consecutive order, starting with the lowest height drop. Do not list the heights at which all the primers tested fired or all the primers tested failed to fire.

Title Column II "Number Fired" and list the number of primers that fired at intermediate height.

Title Column III "Number Misfired" and list the number of primers that failed to fire at each intermediate height.

Title Column IV "Hangfire or blowback" shall be counted as "Misfires" for the purpose of calculation.

Title Column V "Fraction Misfired - P_i" and list the decimal fraction to the closest second decimal place, of the primers that failed to fire at each intermediate height. This number is obtained by dividing the sum of Column III and IV at each intermediate height, by the number of primers tested at that height.

Title Column VI "Standard Deviation Factor - K_i" and list odd numerals in sequence, starting with 1, for as many intermediate heights as are listed in Column I.

Title Column VII "Standard Deviation Factor Times Fraction Misfired - K_iP_i" and list the product obtained for each intermediate height when Column V is multiplied by Column VI.

Apply formulae and solve for $\bar{H}$ and s.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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